

Running Evo 2 7B on Google Cloud and Runpod: cost, GPU memory and cross-hardware agreement of variant scores

Study repository `rewire-bio/evo2-cloud-operations`

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Abstract

Clinical informatics teams evaluating the Evo 2 DNA language model need to know what it costs to run, what hardware it needs, and whether its outputs are stable when the hardware changes. We ran one pinned Evo 2 7B container image, unchanged, on three single-GPU platforms: a Google Cloud `a3-highgpu-1g` Spot VM (H100 80 GB), a Runpod Secure Cloud H100 80 GB pod and a Runpod L40S 48 GB pod. Each platform ran Arc Institute’s forward-pass correctness test, zero-shot scoring of 3,893 BRCA1 single-nucleotide variants with 8,192 bp windows, embedding extraction for an exon classifier, a memory probe over sequence length and a same-machine repeat. All platforms passed the correctness test (loss 0.34766). The two H100s, from different providers, produced bit-identical variant scores and classifier probabilities. The L40S produced different scores: median absolute difference $1.2\text{e-}04$ against the H100, Spearman correlation 0.983, and 2,627 of 3,893 variants moved by more than 1% of the list in rank, while the AUROC for loss-of-function variants was almost unchanged (0.877 against 0.875). Every machine reproduced its own scores exactly. A single forward pass ran out of memory at 131 kb on both the 48 GB and the 80 GB GPU. Scoring cost USD 0.51 (Runpod H100), 0.46 (Runpod L40S, which varied with the host) and 0.80 (Google Cloud H100 Spot) per 1,000 variants at list prices. Zero-shot Evo 2 scores separated BRCA1 loss-of-function variants better than CADD overall (AUROC 0.875 against 0.804), entirely through noncoding variants. These are operational measurements, not a clinical validation.

1 Introduction

Evo 2 is an open DNA language model with 7B and 40B parameter checkpoints and a context of up to 1 Mb [1]. Its authors report strong zero-shot variant-effect results, particularly for noncoding and splice variants, and publish inference code, weights and notebooks [2]. They do not describe how to provision it on commercial clouds, what it costs, or how its outputs behave across GPU types. Users have reported that 7B embeddings differ between H100 and L40S GPUs [3] and that long sequences exhaust GPU memory well short of the advertised context [4]. For a clinical team these are not curiosities: a score that changes with the hardware cannot be locked down by an assay validation, and a context that does not fit is not a usable context.

This study runs one fixed workload on several platforms with the same container image, and records correctness, agreement, memory, time and cost. It also reproduces two of Arc’s published workflows (BRCA1 zero-shot scoring and exon classification) with the 7B checkpoints, against labels and simple baselines. The design, platforms and budget were approved in a written protocol before any GPU run, with three dated amendments.

2 Methods

2.1 Platforms

Table 1 lists the platforms. All are single-GPU machines. Google Cloud’s `a3-highgpu-1g` is sold only as Spot or Flex-start; we used Spot. Runpod pods ran in Secure Cloud, on demand. The

Google Cloud H100 is the reference platform for agreement comparisons. AWS `p5.4xlarge` and `g6e.2xlarge` launchers are included in the repository but were not run (no account was available; amendment 2). Runpod chose the data centre for each pod; we did not constrain it.

2.2 Software and model pins

The container is built from `nvcr.io/nvidia/pytorch:25.04-py3` (the base of Arc’s Dockerfile), pinned by digest, with `evo2==0.6.0` and `vtx==1.1.0`. It contains PyTorch 2.7.0 (NVIDIA build), flash-attn 2.7.3 and Transformer Engine; its full package list is recorded. The image is referenced by digest on every platform. Weights were downloaded from Hugging Face at pinned revisions and checked against SHA-256 hashes before loading, because Arc’s loader uses `torch.load`, which can execute code. The exon classifier (`schmojo/evo2-exon-classifier`) is a two-layer network; instead of loading it with `trust_remote_code` as Arc’s notebook does, we rebuilt the network locally and loaded only the pinned `safetensors` weights. Study code was copied to each machine from a committed revision; all three platforms in the recorded run used the same revision.

2.3 Data

All inputs are public and were fetched from Arc’s repository at a pinned commit, with SHA-256 checks: GRCh37 chromosome 17; the Findlay et al. BRCA1 saturation genome editing table [5] (3,893 SNVs: 2,821 functional, 249 intermediate, 823 loss of function, with consequence, CADD and mammalian phyloP columns); 122 labelled exon and non-exon positions; and Arc’s four forward-pass test sequences. No patient data were used.

2.4 Workloads

W0 reimplements Arc’s forward-pass test with `evo2_7b`; a platform passes if the mean loss is within 0.001 of Arc’s expected 0.3477. **W1** scores 8,192 bp windows centred on each BRCA1 SNV, as in Arc’s notebook: the score is the variant window’s mean log-likelihood minus the reference window’s (batch size 1). The 3,893 variants share 1,326 reference windows, so 5,219 sequences are scored. **W2** extracts final-token embeddings from layer `blocks.26` of `evo2_7b_base` for forward and reverse 8,192 bp sequences at 122 positions and scores them with the pinned classifier. **W3** runs one forward pass of `evo2_7b` on synthetic sequences of 8,192, 32,768, 131,072 and 262,144 bp and stops at the first out-of-memory error. **W4** rescores the first 200 BRCA1 variants in a new process on the same machine.

2.5 Metrics

Discrimination is the AUROC for loss-of-function against functional or intermediate variants, using the negated W1 score, with 95% intervals from 2,000 bootstrap resamples of positions (the three SNVs at a position are kept together). Baselines from the same table need no GPU: CADD, mammalian phyloP (amendment 1 corrected its sign before any Evo 2 output existed) and a consequence rule (nonsense and canonical splice, then splice region, then missense, then other). Paired differences use the same resamples. Agreement between platforms is the maximum and median absolute difference in per-variant scores, the Spearman correlation, and the number of variants whose rank moves by more than 1% of the list (39 places). Peak memory is `torch.cuda.max_memory_allocated`, with `nvidia-smi` sampled every 500 ms. Cost is billed time (from pod creation on Runpod, from `RUNNING` on Google Cloud, to deletion) times the list price retrieved by API at run time, plus disk.

3 Results

3.1 Correctness and agreement

All three platforms passed W0 with mean loss 0.34766. Each platform reproduced its own W4 scores exactly. The Runpod H100 matched the Google Cloud H100 exactly on all 3,893 variant scores and all 122 classifier probabilities (maximum absolute difference 0). The L40S did not (Figure 1): its scores differed from the reference by a median of $1.2\text{e-}04$ and at most $1.3\text{e-}03$, with Spearman correlation 0.983; 2,627 variants moved by more than 39 rank places, and classifier probabilities differed by up to 0.020. For scale, the BRCA1 variant scores themselves are of order 10^{-4} to 10^{-2} . Two earlier runs of all three platforms on fresh machines, on the same day, gave BRCA1 scores bit-identical to the recorded run on every platform (run log).

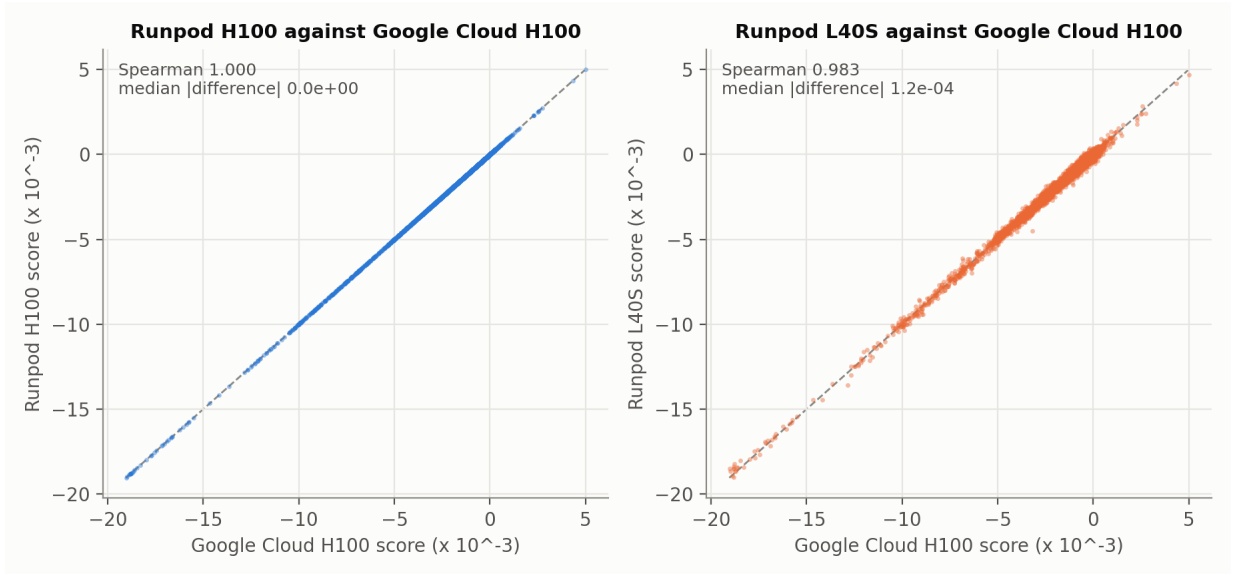


Figure 1: Per-variant BRCA1 scores on each platform against the Google Cloud H100. Same image, inputs and code. Left: Runpod H100 (identical). Right: Runpod L40S. Dashed line: identity.

3.2 BRCA1 discrimination and baselines

Zero-shot Evo 2 7B scores gave AUROC 0.875 [0.859, 0.891] for all SNVs, against 0.804 for CADD, 0.781 for phyloP and 0.769 for the consequence rule (Figure 2). The difference from CADD, +0.071 [+0.050, +0.095] in CADD’s disfavour, came from noncoding variants (Evo 2 0.953 against CADD 0.875; difference +0.077 [+0.044, +0.113]). On coding variants the two were level (Evo 2 0.840, CADD 0.841; difference -0.001 [-0.026, +0.023]). The exon classifier gave AUROC 0.945 on 122 positions; its training data may include these positions, so this is a pipeline check rather than an evaluation.

3.3 Memory

Scoring 8,192 bp windows needed 16.8 GiB allocated by PyTorch on every platform. A single 32,768 bp forward pass needed 30.4 GiB. At 131,072 bp the forward pass ran out of memory on both the 80 GB H100 and the 48 GB L40S (Figure 3). The checkpoint’s 1 Mb context is therefore not reachable in one forward pass on a single GPU with Arc’s inference code; Arc points to multi-GPU frameworks for long sequences [4].

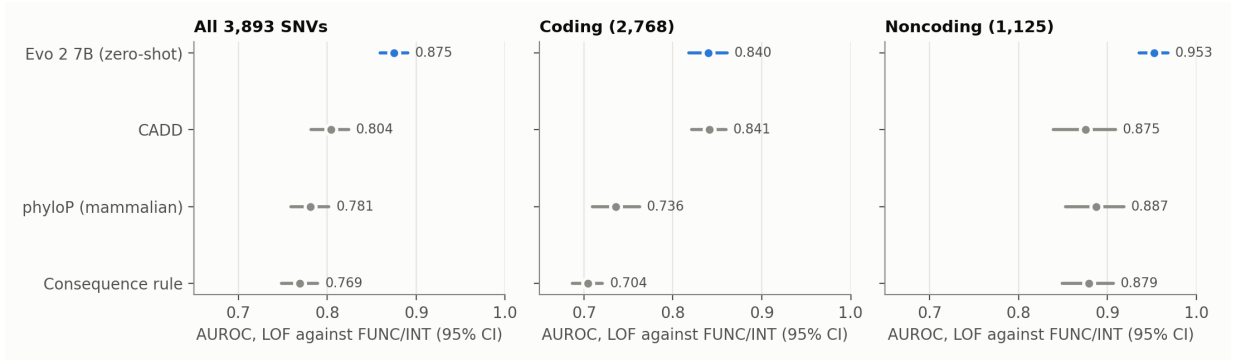


Figure 2: AUROC for BRCA1 loss-of-function variants with 95% grouped bootstrap intervals, reference platform. Baseline scores come from the Findlay et al. table.

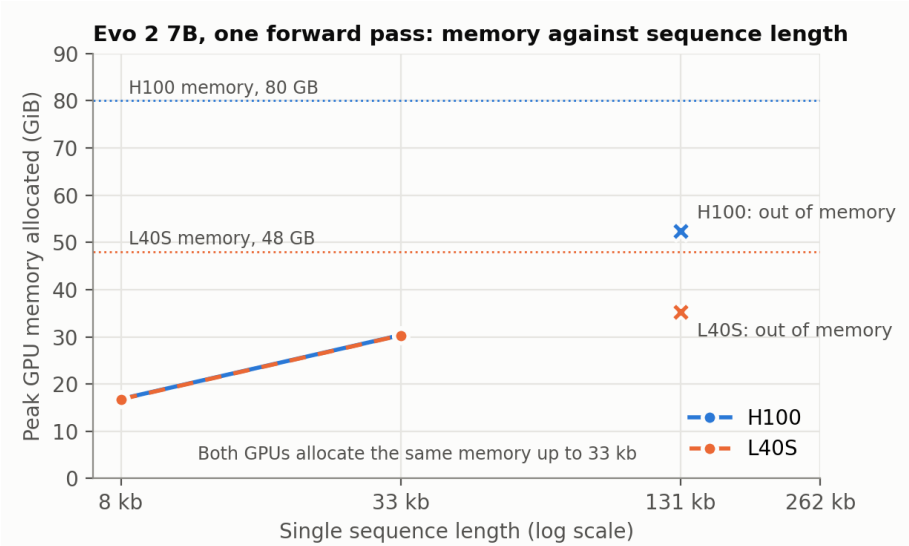


Figure 3: Peak GPU memory for one forward pass against sequence length. Crosses mark the memory in use when the next allocation failed.

Table 1: Platforms and measured operations for the recorded run. USD per 1,000 variants assumes the BRCA1 pattern of 5,219 scored windows per 3,893 variants and list prices at run time.

Platform	GPU	Location	Purchase	USD/h	Windows/s	USD/1k var.	Minutes	USD
Google Cloud H100	H100	us-central1-c	spot	6.29	2.92	0.80	44	4.62
Runpod H100	H100	AP-IN-1	on-demand	4.02	2.92	0.51	41	2.74
Runpod L40S	L40S	US-MO-1	on-demand	1.12	0.91	0.46	113	2.11

3.4 Throughput and cost

The two H100s scored at 2.92 and 2.92 windows per second and the L40S at 0.91 (Table 1). L40S throughput depended on the host: an L40S used earlier the same day scored 0.93 windows per second (run log). Its cost per variant therefore ranged from below to above that of the Runpod H100, depending on the host it was given. The Google Cloud Spot H100 cost more per variant than the Runpod on-demand H100, because its hourly price was higher at the same throughput. Fixed setup costs were large relative to this short workload: pulling the 12 GB compressed image took 393 s on Google Cloud, and Google’s current Deep Learning VM image required installing Docker and the NVIDIA container toolkit first (0 s). The recorded run cost USD 9.47 in total.

3.5 Operational events

The run log records every attempt. Earlier, phased runs on the same three platforms found and fixed four launcher problems before the recorded run: Runpod’s REST API v1 had been superseded by v2; a Runpod account without credit was refused with HTTP 402; Google’s CUDA 12.9 Deep Learning VM image ships without Docker; and it holds the NVIDIA container toolkit at an older version, so the current package cannot be installed alongside it. Runpod placed one H100 pod in India and the L40S in Texas, because no data centre was specified. In the recorded run, the first Google Cloud attempt of the day had lost GPU access for new processes inside the running container, probably because a systemd reload on the host revoked the container’s device permissions; attaching the GPU through CDI and pausing automatic updates resolved it. One L40S attempt failed when the SSH session from the coordinating laptop dropped, because the workload ran in the foreground of that session. L40S throughput varied by host: 0.93 windows per second on the first host and 0.91 on the host used in the recorded run.

4 Discussion

Hardware is part of the method. With the same image, inputs and GPU model, Evo 2 7B outputs reproduced exactly across two providers and across processes. Changing the GPU model changed individual scores by about as much as the scores themselves, reordering most of a ranked list, while leaving aggregate discrimination intact. The 7B checkpoint uses FP8 input projections when Transformer Engine is present, and FP8 and bf16 kernels differ between GPU generations, which is a plausible cause; we did not isolate it. In practice a validated pipeline must fix the GPU model or revalidate when it changes, and Arc’s correctness test is not sufficient on its own: the L40S passed it.

Context capacity is not memory fit. Window lengths beyond about 32 kb need multi-GPU inference. Pipelines that rely on long context, such as EVEE’s 65 kb windows over four million variants (about 20,000 H100-hours) [6], are a different operational class from the 8 kb scoring measured here.

Price and residency. A cheaper 48 GB GPU is adequate for 8 kb scoring, but on a GPU cloud its cost per variant depends on the host it lands on, so measure throughput before committing to a large job. Spot pricing is not automatically cheaper than a GPU cloud’s on-demand price. GPU clouds choose a location unless told otherwise; with identifiable data the location must be pinned, and covered by a business associate agreement where HIPAA applies. EC2, EBS and Compute Engine are covered by their providers’ BAAs [7, 8]; Runpod states HIPAA and GDPR programmes [9]. NVIDIA’s hosted Evo 2 API prohibits protected health information and production use [10].

5 Limitations

- Few runs per platform; timings are receipts, not benchmarks, and throughput varied between hosts with the same GPU model. List prices change.
- Three platforms, two GPU models. AWS was not run. Blackwell GPUs, where a user reports wrong losses [11], were not tested.
- Only the 7B checkpoints. The 40B model needs two H100s or one H200 [12].
- The BRCA1 comparison is one gene from one assay; the exon task may overlap the classifier’s training data. Neither says anything about clinical validity, calibration or ancestry.
- The cause of the L40S differences was not isolated (for example, by disabling FP8).
- Compliance terms were taken from provider pages, not audited.

6 Availability

Code, protocol, amendments, run log, results and this manuscript: <https://github.com/rewire-bio/evo2-cloud-operations>. Container image: `ghcr.io/rewire-bio/evo2-cloud-operations` (digest in `configs/full.json`).

References

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